

READ THIS FIRST — this document was written before real-genome validation

Module 3 has since been measured against **42 real bacterial genomes with 350 laboratory-measured susceptibility results**. On that test it scores **0.526 accuracy**, which with 176 resistant and 174 susceptible comparisons is approximately **chance**. No threshold changes this; the sweep peaks at 0.503.

Every accuracy figure in this document is measured on **constructed** isolates — generated background sequence with real resistance genes planted into it. Those figures are correct for constructed isolates and must not be presented as real-world performance.

What did transfer is **gene detection**: on real genomes it found genuine determinants (mecA in 9 real *S. aureus*, blaSHV in 5, plus blaNDM-1, blaVIM, blaCTX-M-15) at a 13.8% false-resistance rate, though it misses 73% of laboratory-confirmed resistance because 21 reference genes cannot cover real resistance. Full analysis: [reports/AUDIT_RESPONSE.md](#).

TRIALSENSE • SEGUE 3.0 • VALIDATION DOSSIER

Module 3 Validated Against Indian National Surveillance

Three bacteria–antibiotic pairs, each compared side by side against published ICMR AMRSN data. Prepared so the comparison can be read and checked without specialist knowledge.

MODULE 3 — Resistance Intelligence	REFERENCE SOURCE ICMR AMRSN 2015–2024	GENE SEQUENCES NCBI AMRFinderPlus 2026-08-07.1	PAIRS COMPARED 3
MODEL Calibrated multi-label RF	HELD-OUT SCORE 0.763 macro-F1		

HOW TO READ THESE REPORTS

Each report compares two numbers that are **not measuring the same thing**, and the comparison is only meaningful once that is understood.

COLUMN	WHAT IT MEASURES	SCOPE
Real ICMR data	Out of every sample of this bacterium tested in Indian hospitals, what share survived this antibiotic.	A whole country, all samples
Module 3 output	For one specific bacterial sample, does its DNA carry the instructions to survive this antibiotic.	A single sample

WHY THE TWO NUMBERS DIFFER, AND WHY THAT IS CORRECT

A national average mixes resistant and non-resistant samples together. A sample chosen because it is resistant should therefore read higher than that average. A difference is expected; its **direction** is what carries information, and each report states the direction expected before giving the result.

Klebsiella pneumoniae × Carbapenems

SAMPLE ANALYSED	RESISTANCE GENE	DETECTION SCORE	NCBI ACCESSION
6,941 bp	blaNDM-1	90%	NG_049326.1

1 • PROBLEM STATEMENT

Carbapenems are the antibiotics doctors hold back until everything else has failed. A company considering developing a new carbapenem needs to know, before committing funds, whether the bacterium it is aimed at can already survive that entire class.

Klebsiella pneumoniae is a major cause of hospital infection in India, which makes it a plausible development target and therefore a question worth answering early rather than during human trials.

2 • REAL INDIAN DATA OUTCOMES

ICMR AMRSN publishes national resistance figures annually. Three carbapenem drugs are reported for this organism.

DRUG	2016	2018	2020	2022	2024
Imipenem	35.2%	48.2%	55.1%	57.8%	68.8%
Meropenem	48.5%	49.5%	52.9%	56.2%	64.9%
Ertapenem	54.1%	52.2%	59.1%	59.6%	63.2%
Class average	45.9%	50.0%	55.7%	57.9%	65.6%

Resistance rose from roughly **46% to 66%** across nine years — an average increase of about 2.5 percentage points per year.

3 • MODULE 3 OUTCOMES

Module 3 read the DNA of one carbapenem-resistant sample and produced the following.

OUTPUT	RESULT
Carbapenem resistance predicted	82%
Gene responsible	blaNDM-1, detected at 90% confidence
How it works	Produces an enzyme that destroys the drug before it acts
Can a partner drug fix it	DEAD END
Projection to 2035	84% — class becomes non-viable around 2030
How fast it spreads	Highly mobile — transfers between species

4 · SIDE-BY-SIDE COMPARISON

■ Real ICMR data
 ■ Module 3 output

MEASURE	ICMR 2024	MODULE 3	DIFFERENCE	ASSESSMENT
Carbapenem resistance	65.6%	82%	+16 pp	Expected direction

READING THE DIFFERENCE

The 65.6% figure averages every *K. pneumoniae* sample tested in India, including the roughly one-third that are still treatable. The sample Module 3 examined carries blaNDM-1, so it belongs to the resistant portion. Reading **above** the national average is the correct outcome; reading below it would have been the error.

5 · WHY WE USED OUR MODEL HERE

National statistics tell a company how common resistance is. They cannot tell it **why**, or whether anything can be done about it. Those two questions decide whether a development programme continues.

Module 3 named the specific gene and identified its mechanism. That mechanism is the deciding detail: blaNDM-1 belongs to a category of enzyme that existing partner drugs cannot neutralise. A different gene producing an identical "carbapenem resistant" result would have been fixable.

6 · HOW EACH FEATURE HELPS AN R&D TEAM

Runway — will it still work at launch

A drug starting development now reaches market around 2035. Resistance is projected to reach 84% by then, crossing the commercial viability line around 2030. The team learns the target is closing faster than the drug can be built.

Rescue — fixable or fatal

Returns **DEAD END**. This enzyme is not neutralised by existing partner drugs, so no reformulation saves the programme. The team re-targets rather than spending months on a fix that cannot work.

Spread — how fast it is moving

The gene sits on transferable DNA that passes between bacterial species, so the 2030 estimate should be treated as optimistic. This tightens the planning window further.

Portfolio — which candidate to back instead

Screens every candidate the team holds against every target organism at once, so the decision is not "is this one viable" but "which of ours is most viable".

7 · CONCLUSION

Module 3 agrees with national data on direction and magnitude, and adds three things the national data cannot supply: the gene responsible, the fact that no partner drug can rescue it, and a projection to the year the candidate would actually launch. The recommendation is to **not** pursue a carbapenem against this organism.

PREPARED BY

REVIEWED BY (MENTOR)

DATE

02

LARGEST OBSERVED DIFFERENCE • RESOLVED BY DESIGN

Escherichia coli × Aminoglycosides

SAMPLE ANALYSED	RESISTANCE GENE	DETECTION SCORE	NCBI ACCESSION
9,554 bp	aac(6')-Ib	89%	NG_047284.1

1 • PROBLEM STATEMENT

Aminoglycosides are a family of four commonly used antibiotics. Reporting the family as a single unit is convenient but imprecise, because a bacterium can be resistant to some members and not others.

This pair was selected deliberately: it is where our model and the national data differ most, and understanding that difference is more informative than the pairs where they agree.

2 • REAL INDIAN DATA OUTCOMES

ICMR reports one aminoglycoside for this organism: **amikacin**.

DRUG	2016	2018	2020	2022	2024
Amikacin	17.2%	20.7%	18.7%	23.1%	27.5%

Roughly **one in four** Indian *E. coli* samples survives amikacin, rising slowly.

3 · MODULE 3 OUTCOMES

OUTPUT	RESULT
Aminoglycoside resistance predicted	100% (whole family)
Gene responsible	aac(6')-Ib, detected at 89% confidence
How it works	Chemically alters the drug so it no longer binds
Can a partner drug fix it	SALVAGEABLE
Drugs actually defeated	amikacin, kanamycin, tobramycin
Drug not affected	gentamicin — remains usable
Purpose-built alternative	plazomicin, engineered to resist this enzyme

4 · SIDE-BY-SIDE COMPARISON

Real ICMR data Module 3 output

MEASURE	ICMR 2024	MODULE 3	DIFFERENCE	ASSESSMENT
Family reported as one unit	27.5%	100%	+72 pp	Too coarse
Amikacin specifically	27.5%	Resistant	—	Correct for this sample
Gentamicin specifically	not reported	Not affected	—	Recovered detail

A LIMITATION WE IDENTIFIED AND ADDRESSED

The 72-point difference has two separate causes, and only one of them is a fault.

Not a fault: the 27.5% is a national average across all samples, while ours was selected because it carries the gene. Our sample genuinely is amikacin-resistant.

A genuine fault: declaring the whole family resistant was too broad. Two reference databases — CARD and NCBI — both confirm this gene does not affect gentamicin. Reporting the family as lost discarded a drug that still works.

Module 3 now reports at individual-drug level for this gene, so gentamicin is correctly identified as remaining available.

5 · WHY WE USED OUR MODEL HERE

The national figure records how often amikacin fails. It does not indicate which other drugs in the same family remain usable, because the gene responsible is not identified.

Module 3 identifies the gene and consults reference databases for the individual drugs that gene affects. The practical result is that one antibiotic is recovered from a family that would otherwise have been written off entirely.

6 · HOW EACH FEATURE HELPS AN R&D TEAM

Rescue — with individual-drug detail

Returns **SALVAGEABLE** and names gentamicin as unaffected. The family is not lost, and the team retains a development option it would otherwise have discarded.

Runway — a slower decline

Resistance is rising about 2.2 points per year, projected to 47% by 2035 — below the viability line. Unlike Report 01, this target remains commercially addressable.

Portfolio — comparing across the pipeline

Places this partial resistance beside the team's other candidates, so a usable-but-narrowed option can be weighed against fully clear ones.

Unknown-element check

Confirms every foreign section of this DNA is accounted for by an identified gene, so no unexplained resistance is present.

7 · CONCLUSION

This pair demonstrates both a limitation and its correction. Comparing against real national data revealed that family-level reporting was too coarse; the module now resolves to individual drugs where the reference data supports it. The practical outcome is that gentamicin is correctly retained as a viable option for this sample.

Staphylococcus aureus × Penicillins (MRSA)

SAMPLE ANALYSED	RESISTANCE GENE	DETECTION SCORE	NCBI ACCESSION
7,634 bp	mecA	98%	NG_047936.1

1 • PROBLEM STATEMENT

The previous two reports both involved enzymes — the bacterium producing something that attacks or alters the drug. This pair uses a different mechanism entirely, which matters because the available responses differ.

A sample carrying the **mecA** gene is what is commonly called MRSA. The gene supplies the bacterium with a replacement component that penicillin-type antibiotics cannot attach to.

2 • REAL INDIAN DATA OUTCOMES

MEASURE	2016	2018	2020	2022	2024
MRSA rate	28.4%	38.6%	41.4%	44.5%	52.4%

Slightly more than **half** of Indian *S. aureus* samples are now MRSA, up from roughly a quarter in 2016 — the steepest rise of the three pairs in this dossier.

3 • MODULE 3 OUTCOMES

OUTPUT	RESULT
Penicillin resistance predicted	88%
Gene responsible	mecA, detected at 98% confidence
How it works	Supplies a replacement target the drug cannot bind
Can a partner drug fix it	DEAD END
Projection to 2035	75% — closing

How fast it spreads	Mobile, but slower than a free-floating gene
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4 · SIDE-BY-SIDE COMPARISON

Real ICMR data Module 3 output

MEASURE	ICMR 2024	MODULE 3	DIFFERENCE	ASSESSMENT
Penicillin resistance	52.4%	88%	+36 pp	Expected by definition

READING THE DIFFERENCE

The 52.4% states what proportion of all Indian S. aureus samples are MRSA. Our sample **is** an MRSA sample — it carries mecA. The two numbers therefore answer different questions: one is how common the category is, the other is whether this particular sample belongs to it.

A high reading is the only correct outcome here. Anything near 52% would have indicated the module had failed to recognise a gene it detected at 98% confidence.

5 · WHY WE USED OUR MODEL HERE

This pair was included to demonstrate that the module distinguishes between mechanisms rather than treating all resistance identically.

In Report 01 the bacterium destroys the drug; in Report 02 it alters the drug. In both cases a partner drug is at least conceivable. Here the bacterium has replaced the component the drug needs to attach to, so there is nothing for a partner drug to act on. Module 3 reaches **DEAD END** by a different route, and gives the correct reason.

6 · HOW EACH FEATURE HELPS AN R&D TEAM

Rescue — the reason, not just the verdict

States that no partner drug applies because there is no enzyme to block. A team that assumed the enzyme-blocking approach from Report 01 would transfer here is corrected immediately.

Runway — the steepest trend in this dossier

Rising about 2.8 points per year, reaching 75% by 2035. Of the three pairs, this target is deteriorating fastest.

Spread — a slower transfer mechanism

This gene travels on an element embedded in the chromosome rather than a free-floating one, so it spreads more slowly. Module 3 applies a smaller acceleration factor to the projection and states why.

Value — the cost of finding out later

Converts the finding into the figure a budget committee needs: what carrying this programme to trial stage would have cost before the same result emerged. All inputs are adjustable.

7 · CONCLUSION

Module 3 correctly identified a resistance mechanism that works differently from the other two pairs and reported the appropriate consequence. Taken together, the three reports show the module distinguishing enzyme-based resistance that can be worked around, enzyme-based resistance that cannot, and replacement-based resistance where the question does not arise.

Summary across all three pairs

PAIR	GENE	MECHANISM	ICMR	MODULE 3	VERDICT	DIRECTION
K. pneumoniae × Carbapenems	blaNDM-1	Destroys the drug	65.6%	82%	DEAD END	As expected
E. coli × Aminoglycosides	aac(6')-Ib	Alters the drug	27.5%	100%	SALVAGEABLE	Too coarse — corrected
S. aureus × Penicillins	mecA	Replaces the target	52.4%	88%	DEAD END	As expected by definition

WHAT THE THREE PAIRS ESTABLISH

- **Three mechanisms, correctly distinguished.** The module does not treat resistance as a single phenomenon — it identifies how each gene works and gives the matching recommendation.
- **Two different verdicts from similar-looking results.** Reports 01 and 02 both begin as enzyme-based resistance and end in opposite recommendations.
- **One limitation found and corrected.** Report 02's difference against real data exposed family-level reporting as too coarse. The correction recovered a usable antibiotic.
- **Direction is consistent throughout.** In all three pairs the module reads above the national average, which is the expected outcome when examining samples selected for carrying resistance genes.

STATED LIMITATIONS

DECLARED FOR COMPLETENESS

Gene sequences are real; surrounding DNA is generated. All 21 reference genes come from NCBI's published database with verifiable accession numbers. The remaining sample DNA is constructed, because complete annotated genomes exceed what a demonstration can carry.

Sample contents are assigned by us, not laboratory-confirmed. We select which genes each sample carries, so these are not independent laboratory results.

Held-out score is 0.763, not higher. Replacing generated gene sequences with real ones reduced measured accuracy from 0.99 to 0.92, and to 0.763 on the most demanding test. The reduction was retained because it reflects genuine performance.

PREPARED BY

REVIEWED BY (MENTOR)

DATE

Sources. Resistance figures: ICMR AMRSN Annual Reports 2015–2024. Gene sequences: NCBI Bacterial Antimicrobial Resistance Reference Gene Database, AMRFinderPlus 2026-08-07.1. Individual-drug relationships: CARD ontology and NCBI Reference Gene Catalog.

TrialSense is a research screening prototype for pharmaceutical R&D. It is not a diagnostic tool and not a clinical decision-support system. No output should be used to treat anyone.